

RAG Knowledge Assistant

Source-grounded AI assistant for document search, cited answers, and retrieval transparency.

[View Live Demo](#)[GitHub Repo](#)

Problem

Knowledge workers often search across long PDFs and internal documents to answer specific questions. Traditional search returns documents, not answers. Generic LLM responses can hallucinate or lack evidence. This product focuses on fast, source-grounded answers users can verify.

Why It Matters

Built for trust-sensitive document workflows where users need citations, source visibility, and honest no-answer behavior instead of unsupported AI guesses.

Product Solution Workflow



Product Highlights

- Multi-PDF upload and ingestion
- Conversational Q&A over indexed documents
- Inline citations with filename, page, and chunk
- Source panels and PDF preview for verification
- Behind-the-scenes retrieval/debug visibility

Product Decisions I Made

Problem Framing

Framed the product around a trust gap: users need verifiable answers from long documents, not another generic chatbot.

User Journey Design

Mapped the workflow from upload -> ingestion -> question -> cited answer -> source inspection to reduce cognitive load.

MVP Prioritization

Prioritized citations, source panels, PDF preview, and no-answer behavior before advanced features.

Trust-First UX

Designed the experience so users can validate the answer, inspect evidence, and understand when the AI lacks support.

AI Evaluation

Created positive, negative, edge-case, and unsupported-question tests to check grounding and hallucination risk.

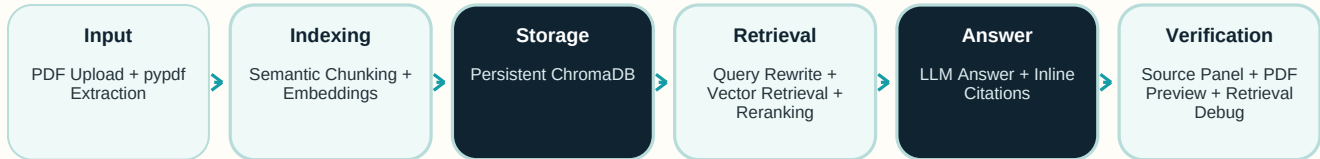
Observability

Added retrieval/debug visibility, source chunks, and token usage to make AI behavior easier to inspect and improve.

Supporting Detail

Architecture, trust, evaluation, observability, stack

Architecture Snapshot



Trust & Evaluation

- Answers grounded in retrieved document chunks
- Citations help users verify claims
- No-answer behavior for unsupported questions
- Edge-case prompts test hallucination risk
- Source panels and PDF preview reduce black-box behavior

UX & Observability

- Ingestion status cards show processing state
- Indexed document views expose stored files and chunks
- Source inspection shows retrieved evidence
- Retrieval/debug panels help diagnose answer quality
- Token usage visibility supports cost awareness

Technical Stack

